

04/04/2024

Phase-I
CODE-A

Aakash

Medical | IIT-JEE | Foundations

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FINAL TEST SERIES for NEET-2024

MM : 720

Test-5

Time : 3 Hrs. 20 Mins.

Topics covered :

Physics : Electric Charges and Fields, Electrostatic Potential and Capacitance

Chemistry : Solutions, Electrochemistry, Chemical Kinetics

Botany : Plant Growth and Development, Sexual Reproduction in Flowering Plants

Zoology : Chemical Coordination and Integration, Human Reproduction

Instructions :

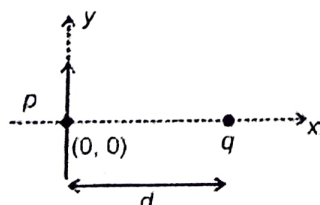
- There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.
- Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

PHYSICS

Choose the correct answer :

SECTION - A

1. A charge q is placed at a distance d from short electric dipole placed along y -axis as shown below.

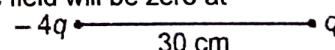
The force on charge q is

- | | |
|--------------------------------|---------------------------------|
| (1) $\frac{2Kqp}{d^3} \hat{i}$ | (2) $-\frac{2Kpq}{d^3} \hat{j}$ |
| (3) $\frac{Kpq}{d^3} \hat{i}$ | (4) $-\frac{Kpq}{d^3} \hat{j}$ |

2. Two identically charged conducting spheres P and Q are separated by distance d . The force between them is F . Now another identical uncharged conducting sphere first touched with P and then with Q and finally removed. The new force between P and Q is

- | | |
|--------------------|--------------------|
| (1) $\frac{F}{2}$ | (2) $\frac{F}{4}$ |
| (3) $\frac{3F}{4}$ | (4) $\frac{3F}{8}$ |

3. Two charges $-4q$ and q are 30 cm apart. The electric field will be zero at

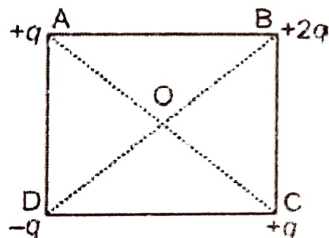


- | | |
|--------------------|--------------------|
| (1) 30 cm from q | (2) 60 cm from q |
| (3) 15 cm from q | (4) 10 cm from q |

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4. If electric field $\vec{E}$ is zero in a region, then electric potential V in that region
- Must be zero
 - Must be non-zero
 - Must be constant
 - Both (2) and (3)
5. Consider the following statements :
- The electric field due to charge outside the Gaussian surface contributes zero net flux through the surface
 - Total flux linked with a closed surface, not enclosing any charge is zero
 - Total electric flux, if a dipole is enclosed by surface is zero
- The correct statement(s) is/are
- Only (a) and (b)
 - Only (b) and (c)
 - Only (a) and (c)
 - (a), (b) and (c)
6. Four charges are arranged at the corners of a square ABCD, as shown in the figure. The direction of force on the negative charge kept at the centre O is



- Net force is zero
 - Along O to D
 - Along O to B
 - Along O to C
7. **Assertion (A):** On disturbing an electric dipole from unstable equilibrium in an electric field, it return back to its unstable equilibrium orientation.
Reason (R): A restoring torque acts on the dipole on being disturbed from its unstable equilibrium.
- Both (A) and (R) are true and (R) is the correct explanation of (A)
 - Both (A) and (R) are true and (R) is not the correct explanation of (A)
 - (A) is true but (R) is false
 - Both (A) and (R) are false statements

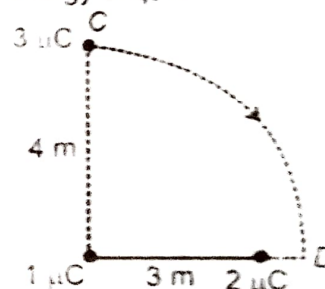
8. A charge Q is given to a parallel plate capacitor. The energy density between the plates is

- $\frac{1}{2} \frac{Q^2}{\epsilon_0 A^2}$
- $\frac{Q^2}{\epsilon_0 A^2}$
- $\frac{2Q^2}{\epsilon_0 A^2}$
- Zero

9. A hollow metal sphere of radius 5 cm is charged such that potential at surface is 10 V. The potential at a distance 10 cm from the centre will be

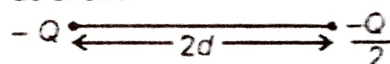
- 5 V
- $\frac{5}{2}$ V
- $\frac{5}{4}$ V
- Zero

10. Two charges $q_1 = 1 \mu\text{C}$ and $q_2 = 2 \mu\text{C}$ are kept at 3 m apart, as shown in the figure. A third charge $q_0 = 3 \mu\text{C}$ is moved along the arc of circle of radius 4 m from C to D as shown. The change in potential energy of q_0 is



- 51.4 mJ
- 43.2 mJ
- 2.1 mJ
- 27.1 mJ

11. Two charges $-Q$ and $-\frac{Q}{2}$ are placed along a line as shown



The system of charges will have potential energy equal to

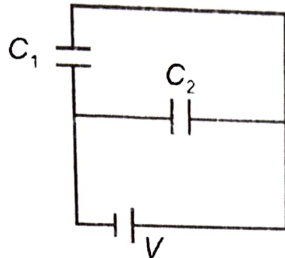
- $\frac{KQ^2}{4d}$
- $\frac{-KQ^2}{4d}$
- $\frac{-5KQ^2}{4d}$
- $\frac{-5KQ^2}{d}$

Space for Rough Work

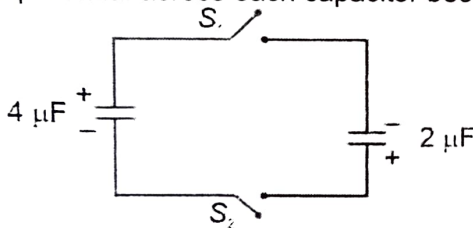
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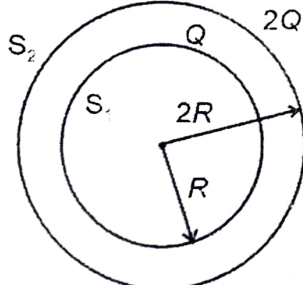
12. A network of two capacitors of capacity equal to $C_1 = C$ and $C_2 = 3C$ are connected to a battery as shown in the figure. The ratio of energy stored in C_1 to C_2 is



- (1) $\frac{1}{3}$ (2) $\frac{1}{9}$
(3) $\frac{1}{54}$ (4) $\frac{1}{36}$
13. Two capacitors of capacity $2 \mu\text{F}$ and $4 \mu\text{F}$ are charged to same potential difference of 4 V . Now they are connected with opposite polarity as shown. After closing the switches S_1 and S_2 , the final potential across each capacitor become



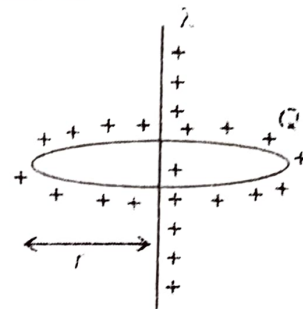
- (1) Zero (2) $\frac{2}{3} \text{ V}$
(3) $\frac{4}{3} \text{ V}$ (4) 3 V
14. S_1 and S_2 are two hollow concentric conducting spheres having charges Q and $2Q$ respectively.



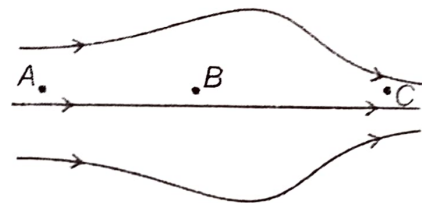
The ratio of electric field on the surface S_1 to surface S_2 respectively is

- (1) $\frac{1}{2}$ (2) $\frac{1}{4}$
(3) $\frac{2}{3}$ (4) $\frac{4}{3}$

15. An infinitely long uniformly charged wire is placed along the axis of a uniformly charged ring. The linear charge density on wire is λ and charge Q is uniformly distributed over ring. The electric force between them is



- (1) $\frac{\lambda Q}{2\pi\epsilon_0 r}$ (2) $\frac{\lambda Q}{4\pi\epsilon_0 r}$
(3) $\frac{\lambda Q}{\pi\epsilon_0 r}$ (4) Zero
16. The electric field lines in a region is shown in the figure. If E_A , E_B and E_C are the electric field strength at points A, B and C respectively, then



- (1) $E_A = E_B = E_C$ (2) $E_C > E_A > E_B$
(3) $E_A > E_C = E_B$ (4) $E_B > E_A > E_C$
17. There is an electric field E in x -direction. If the magnitude of work done on moving a charge $2 \mu\text{C}$ through a distance of 2 m along a line making angle 60° with x -axis is 4 mJ , the electric field E is
- (1) $2 \times 10^3 \text{ N/C}$ (2) 10^3 N/C
(3) $4 \times 10^3 \text{ N/C}$ (4) 400 N/C

Space for Rough Work

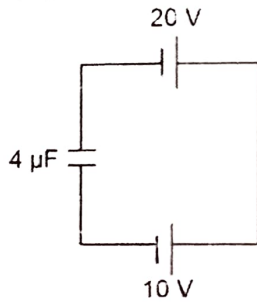
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18. Two capacitors C_1 and C_2 are charged to 40 V and 120 V respectively. It is found that by connecting them together the potential across each can be made zero. Then $\frac{C_1}{C_2}$ is equal to

- (1) 3 (2) $\frac{1}{3}$
(3) $\frac{2}{3}$ (4) $\frac{3}{2}$

19. In the given circuit, charge stored on $4 \mu\text{F}$ capacitor will be



- (1) $120 \mu\text{C}$ (2) $40 \mu\text{C}$
(3) $10 \mu\text{C}$ (4) $30 \mu\text{C}$

20. A $20 \mu\text{F}$ capacitor is charged to 1 kV potential and then its plates are connected by a conducting wire. How much heat would be produced?

- (1) 5 J (2) 10 J
(3) 15 J (4) 17.5 J

21. Three capacitors of capacitance $18 \mu\text{F}$, $9 \mu\text{F}$ and $3 \mu\text{F}$ are connected once in series and then in parallel. The ratio of equivalent capacitance in

two cases $\frac{C_{\text{parallel}}}{C_{\text{series}}}$ is

- (1) 9 : 1 (2) 6 : 1
(3) 15 : 1 (4) 9 : 4

22. A parallel plate capacitor with air between the plates has capacitance of 8 pF. What will be capacitance if the distance between plates is

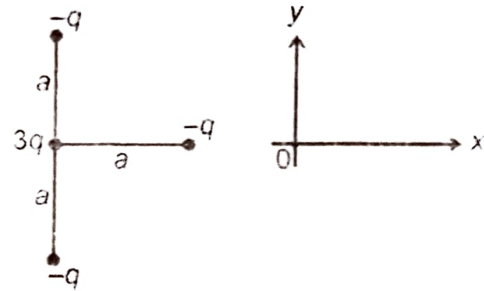
reduced by half and space between them is filled with a material of dielectric constant 6?

- (1) 32 pF (2) 64 pF
(3) 96 pF (4) 16 pF

23. If $\vec{E}_1$ and $\vec{E}_2$ be respective electric fields at equatorial position and axial position of an electric dipole at same distance, then

- (1) $\vec{E}_1 \cdot \vec{E}_2 = 0$ (2) $|\vec{E}_1 + \vec{E}_2| = 0$
(3) $|\vec{E}_1 \times \vec{E}_2| = 0$ (4) $|\vec{E}_1 - \vec{E}_2| = 0$

24. Net dipole moment of the arrangement of the charges as shown in figure is



- (1) $\sqrt{2}qa\hat{i}$ (2) $-qa\hat{i}$
(3) $-\sqrt{2}qa\hat{i}$ (4) $qa\hat{i}$

25. If charge on a body is 2 nC, then how many electrons are absent from the body?

- (1) 1.5×10^{27} (2) 1.25×10^{28}
(3) 1.25×10^{10} (4) 1.6×10^{19}

26. A point charge $8.85 \times 10^{-12} \text{ C}$ is placed at one corner of a cube. The electric flux passing through the cube is

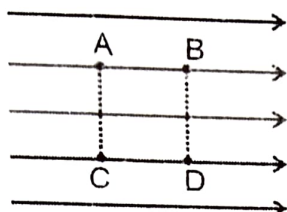
- (1) $1 \text{ N m}^2 \text{ C}^{-1}$ (2) $\frac{1}{6} \text{ N m}^2 \text{ C}^{-1}$
(3) $\frac{1}{8} \text{ N m}^2 \text{ C}^{-1}$ (4) $\frac{1}{24} \text{ N m}^2 \text{ C}^{-1}$

Space for Rough Work

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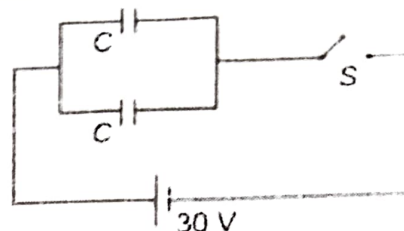
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27. A uniform electric field ($\vec{E}$) is shown in the space given below



The potential difference is zero between points

- (1) A and B (2) A and C
(3) B and C (4) A and D
28. The electric potential V is given as a function of coordinate x (in m) by $V = (5x^2 + 10x - 9)$ volt. The value of electric field at $x = 1$ m is
(1) -20 V/m (2) 6 V/m
(3) -12 V/m (4) 10 V/m
29. If a charged conducting shell has potential V at its centre. What is electric potential at a point 30 cm away from its surface, if radius of shell is 10 cm?
(1) $3V$ (2) $2V$
(3) $\frac{V}{4}$ (4) $\frac{V}{2}$
30. The electric potential at a point due to electric dipole will be maximum and minimum when angles between the axis of dipole and line joining the point to the dipole axis are respectively
(1) 90° and 45° (2) 0° and 90°
(3) 0° and 45° (4) 0° and 180°
31. Minimum number of capacitors each of $8 \mu\text{F}$ and 250 V which can be used to make a composite capacitor of $8 \mu\text{F}$ and 1000 V is
(1) 6 (2) 8
(3) 16 (4) 32
32. Each capacitor shown in figure has a capacitance of $5 \mu\text{F}$. The e.m.f of battery is 30 V and switch S is initially open. The charge that will pass through the battery when switch S is closed, is



- (1) $200 \mu\text{C}$ (2) $300 \mu\text{C}$
(3) $50 \mu\text{C}$ (4) $100 \mu\text{C}$
33. Two capacitors of capacitance $6 \mu\text{F}$ and $12 \mu\text{F}$, each rated to 200 V are connected in series. The maximum voltage that can be applied to the series combination of the given capacitors is
(1) 200 V (2) 400 V
(3) 300 V (4) 600 V
34. Capacitance of a capacitor depends on
(1) Potential difference applied
(2) Charge given to the capacitor
(3) Dimensions of the capacitor
(4) All of these
35. Two charges -1 C and $+1$ C are placed 10 cm apart. Magnitude of electric field at the center of the line joining the two charges is
(1) Zero (2) 9×10^{13} N/C
(3) 72×10^{11} N/C (4) 18×10^{11} N/C

SECTION - B

36. **Assertion (A)** : Electric field inside uniformly charged conducting shell is zero.
Reason (R) : The potential inside a charged conducting shell is variable.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

Test-5 (Code-A)

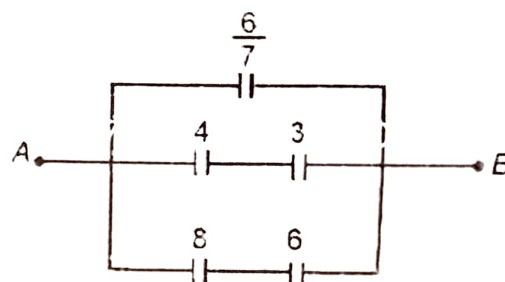
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37. Electric field inside a uniformly charged solid conducting sphere
- (1) Increases from centre to surface
 - (2) Is finite and constant
 - (3) Is zero
 - (4) Decreases from centre to surface
38. **Statement I:** The charge density is smaller on sharp points of an isolated conductor as compared to other parts.
- Statement II:** Equipotential surfaces for a dipole never intersect each other.
- (1) Both statements are correct
 - (2) Both statements are incorrect
 - (3) Statement I is correct and statement II is incorrect
 - (4) Statement I is incorrect and statement II is correct
39. A dielectric slab ($k = 2$) is inserted inside a capacitor connected with a battery as shown in the circuit below. A long time after inserting the slab, match the quantities in column I with their values in column II.

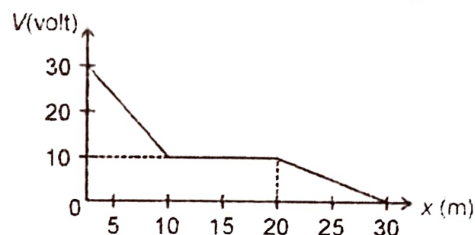


	Column I		Column II
(a)	Charge on capacitor	(P)	4 times of its initial value
(b)	Potential difference across capacitor	(Q)	2 times of its initial value
(c)	Energy of capacitor	(R)	Remain unchanged
(d)	Force between the plates of capacitor		

- (1) (a)-(Q), (b)-(R), (c)-(Q), (d)-(P)
 - (2) (a)-(Q), (b)-(R), (c)-(P), (d)-(Q)
 - (3) (a)-(R), (b)-(R), (c)-(P), (d)-(Q)
 - (4) (a)-(P), (b)-(R), (c)-(Q), (d)-(Q)
40. In the circuit diagram, if all the capacitors are in μF , then the equivalent capacitance between AB is



- (1) $\frac{36}{7} \mu\text{F}$
 - (2) $\frac{48}{7} \mu\text{F}$
 - (3) $6 \mu\text{F}$
 - (4) $9 \mu\text{F}$
41. If 100 J of work must be done to move an electric charge of -2 C from a point where potential is -10 V to another point where the potential is $x \text{ V}$, then value of x is
- (1) 40 V
 - (2) 60 V
 - (3) -40 V
 - (4) -60 V
42. The variation of electric potential with distance x from a fixed point is shown in the figure.



The electric field at $x = 23 \text{ m}$ is

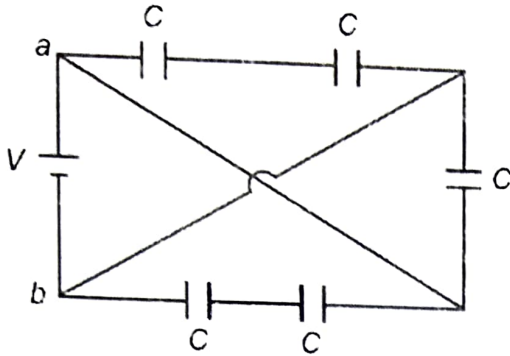
- (1) -1 V/m
- (2) $+1 \text{ V/m}$
- (3) Zero
- (4) -0.5 V/m

Space for Rough Work

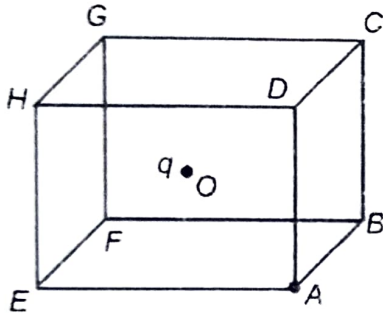
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43. In the diagram shown below, the charge supplied by the battery is

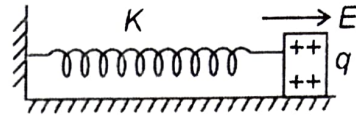


- (1) $\frac{C}{2} V$ (2) $\frac{3}{2} CV$
 (3) $2CV$ (4) $3CV$
44. A charge q is placed at the centre O of a cube of side L as shown. Then the net electric flux through face $ABCD$ is

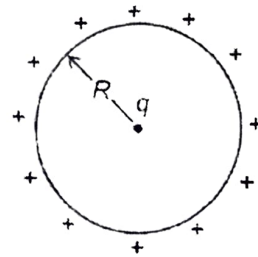


- (1) Zero (2) $\frac{q}{6\epsilon_0}$
 (3) $\frac{q}{3\epsilon_0}$ (4) $\frac{2q}{3\epsilon_0}$
45. An electric dipole is placed in a uniform electric field at unstable equilibrium. Its potential energy is
- (1) Maximum
 (2) Minimum
 (3) Zero
 (4) Half of the maximum value

46. A small block having a charge q is placed on a smooth horizontal table and is connected to a wall through an unstretched spring of spring constant K as shown in figure. A horizontal electric field E parallel to spring is switched on. The amplitude of resulting simple harmonic motion is



- (1) $\frac{qE}{K}$ (2) $\frac{qE}{2K}$
 (3) $\frac{qE}{3K}$ (4) $\frac{2qE}{K}$
47. A thin metallic spherical shell contains a charge Q on it. A point charge q is placed at the centre of the shell. The net force on the charge q is



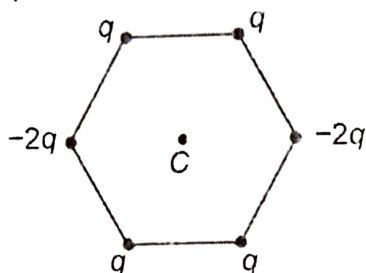
- (1) $\frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2}$ (2) $\frac{1}{\pi\epsilon_0} \frac{qQ}{R^2}$
 (3) $\frac{1}{2\pi\epsilon_0} \frac{qQ}{R^2}$ (4) Zero
48. The dimensional formula for electrical permittivity is
- (1) $[M^{-1}L^{-3}T^4A^{-2}]$
 (2) $[M^{-1}L^{-3}T^{-4}A^2]$
 (3) $[M^{-1}L^{-3}T^4A^2]$
 (4) $[ML^{-3}T^4A^2]$

Space for Rough Work

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49. Six charges are placed at the corners of a regular hexagon as shown in figure. Then choose the correct option among the following



- (1) Both electric field and electric potential will be zero at the centre of the hexagon
(2) Electric field will be zero, but electric potential will not be zero at the centre of the hexagon

- (3) Electric potential will be zero, but electric field will not be zero at the centre of the hexagon
(4) Both electric field and electric potential will be non-zero at the centre of the hexagon

50. The electric field lines are always

- (1) Perpendicular to equipotential surface
(2) Parallel to equipotential surface
(3) Inclined at an angle less than 90° with equipotential surface
(4) Inclined at an angle greater than 90° with equipotential surface

CHEMISTRY

SECTION-A

51. Molarity of a solution obtained after mixing 9 g of glucose in its 250 mL of solution is

- (1) 0.2 M (2) 0.5 M
(3) 2 M (4) 1.5 M

52. The electrode potentials for $\text{Fe}^{3+} + 3\text{e}^- \rightarrow \text{Fe}$ and $\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$ are -0.04 V and -0.44 V respectively. The value of $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}}$ will be

- (1) 0.41 V (2) 0.52 V
(3) 1.00 V (4) 0.76 V

53. Given below are two Statements: One labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): With the increase in temperature, rate of reaction increases.

Reason (R): With the increase in temperature molecules having energy greater than activation energy decreases.

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both (A) and (R) are correct and (R) is not the correct explanation of (A)
(2) (A) is correct but (R) is not correct
(3) (A) is not correct but (R) is correct
(4) Both (A) and (R) are correct but (R) is the correct explanation of (A)

54. If the rate constant for a reaction is $2.303 \times 10^{-2} \text{ s}^{-1}$ then the time required for 90% completion of reaction is

- (1) 75 s
(2) 100 s
(3) 25 s
(4) 50 s

55. The E° value of A^+/A , B^+/B and C^+/C are -0.25 V , $+1.5 \text{ V}$ and -0.85 V respectively. The order of their reducing power is

- (1) $\text{C} > \text{A} > \text{B}$
(2) $\text{A} > \text{B} > \text{C}$
(3) $\text{B} > \text{A} > \text{C}$
(4) $\text{C} > \text{B} > \text{A}$

56. The vapour pressure of solution depends upon
 (i) Temperature
 (ii) Mole fraction of solvent
 (iii) Mole fraction of solute
 (iv) Surface area
 The correct option is
 (1) (i) and (iv) only (2) (i), (ii), (iii) and (iv)
 (3) (ii) and (iii) only (4) (i), (ii) and (iii) only
57. The conductivity of 1.0×10^{-3} M acetic acid is 5×10^{-5} S cm $^{-1}$. The degree of dissociation of acetic acid is
 (If its molar conductivity at infinite dilution is 400 S cm 2 mol $^{-1}$)
 (1) 0.062 (2) 0.125
 (3) 0.50 (4) 0.187
58. van't Hoff factor (i) for acetic acid in water and benzene respectively are
 (1) Greater than unity and greater than unity
 (2) Less than unity and greater than unity
 (3) Greater than unity and less than unity
 (4) Less than unity and less than unity
59. The number of collisions per second per unit volume of the reaction mixture is known as
 (1) Enthalpy
 (2) Activation energy
 (3) Collision frequency
 (4) Rate constant
60. Give below are two statements
Statement-I: In a solution formed by adding carbon disulphide to acetone, the dipolar interactions between solute-solvent molecules are weaker than the respective interaction among solute – solute and solvent-solvent molecules.
Statement-II: In the mixture of phenol and aniline, the intermolecular hydrogen bonding between phenolic proton and lone pair on nitrogen atom of aniline is weaker than the respective intermolecular Hydrogen bonding between similar molecules.

In the light of the above statements, choose the correct answer from the options given below

- (1) Both statement-I and statement-II are false
 (2) Statement-I is true but statement-II is false
 (3) Statement-I is false but statement-II is true
 (4) Both statement-I and statement-II are true
61. The slope of the graph in $\ln k$ versus $\frac{1}{T}$ for Arrhenius equation is
 (1) $\frac{2.303 E_a}{R}$ (2) $\frac{E_a}{R}$
 (3) $\frac{-2.303 E_a}{R}$ (4) $\frac{-E_a}{R}$
62. Reduction potential of hydrogen electrode which is placed in a solution of pH = 4 and $p_{H_2} = 1$ atm, is
 (1) 0.1 V
 (2) -0.24 V
 (3) -0.56 V
 (4) -0.48 V
63. Select the correct statements from the following
 (a) The cell potential for mercury cell remains constant during its life.
 (b) In Leclanche cell Zn container act as cathode.
 (c) A secondary cell can be recharged after use by passing current through it in opposite direction.
- Choose the correct answer from the option given below.
 (1) a and b only
 (2) b and c only
 (3) a and c only
 (4) a, b and c

Test-5 (Code-A)

Final Test Series for NEET-2024

64. Given below are two statements

Statement-I: In collision theory, activation energy and proper orientation of the molecules together determine the criteria for an effective collision.

Statement-II: Collision theory considers atoms/molecules to be hard spheres and ignores their structural aspect

In the light of the above statements, choose the correct answer from the options given below

- (1) Both statement-I and statement-II are false
- (2) Statement-I is true but statement-II is false
- (3) Statement-I is false but statement-II is true
- (4) Both statement-I and statement-II are true

65. Select the correct statements from the following

- (a) Observed colligative property is less than calculated colligative property for a solution of ethanoic acid in benzene.
- (b) For KCl, NaCl and MgSO_4 , value of van't Hoff factor 'i' is close to 2 as the solution becomes very dilute.
- (c) Osmotic pressure is used to determine molar mass of macromolecules like proteins, polymers etc.

Choose the correct answer from the options given below.

- (1) (a) and (b) only (2) (b) and (c) only
- (3) (a) and (c) only (4) (a), (b) and (c)

66. Select the correct statements regarding catalysis from the following

- (a) A small amount of catalyst can catalyse a large amount of reactants.
- (b) A catalyst does not alter Gibbs energy, ΔG of a reaction.
- (c) Catalyst catalyses the spontaneous reactions but does not catalyse non-spontaneous reactions.

Choose the correct answer from the options given below.

- (1) (a) and (b) only (2) (b) and (c) only
- (3) (a) and (c) only (4) (a), (b) and (c)

67. Osmotic pressure of a solution containing 3.42 g of sucrose in 100 mL of solution at 27°C is

- (1) 1.28 atm (2) 2.46 atm
- (3) 4.02 atm (4) 3.1 atm

68. Given below are two statements : One labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): Molecularity greater than three is very rare.

Reason (R): The probability that more than three molecules can collide effectively and react simultaneously is very small.

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both Assertion & Reason are true and the Reason is the correct explanation of the Assertion
- (2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

69. Unit of rate constant (k) for zero order reaction is

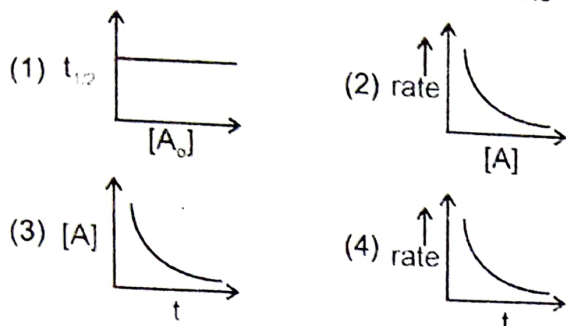
- (1) s^{-1}
- (2) $\text{mol}^{-1} \text{L s}^{-1}$
- (3) $\text{mol L}^{-1} \text{s}^{-1}$
- (4) $\text{mol}^{-2} \text{L}^{-2} \text{s}^{-1}$

Space for Rough Work

Final Test Series for NEET-2024

Test-5 (Code-A)

70. Incorrect graph for a first order reaction is



71. Initial rates of reaction $A + B \rightarrow$ products, are measured for different concentration of reactants A and B as follows

[A]	[B]	rate (Ms^{-1})
0.01 M	0.02 M	1.02×10^{-2}
0.01 M	0.04 M	1.02×10^{-2}
0.02 M	0.04 M	2.04×10^{-2}

The overall order of the reaction is

- (1) 0 (2) 1
 (3) 2 (4) 3

72. Among the following metal ions, which has minimum limiting molar conductivity in aqueous solution at 298 K?

- (1) K^+ (2) Na^+
 (3) Ca^{2+} (4) Mg^{2+}

73. At anode volume of gas obtain at STP during the electrolysis of 0.1M CuSO_4 solution by using 1 A current for 9.65 minute will be

- (1) 33.6 mL (2) 25.2 mL
 (3) 42.7 mL (4) 51.8 mL

74. Unit of cell constant is

- (1) cm (2) cm^{-1}
 (3) cm^2 (4) cm^{-2}

75. Antirust solution among the following is

- (1) Acidic permanganate solution
 (2) Acidic dichromate solution
 (3) Natural magnetic solution
 (4) Alkaline phosphate and alkaline chromate solution

76. If molality of the dilute solution is halved, then the value of molal depression constant will be

- (1) Halved
 (2) Doubled
 (3) Remain same
 (4) Quadrupled

77. Which one of the following electrolytes has the same value of van't Hoff factor (i) as that of $\text{Mg}_3(\text{PO}_4)_2$ (if all are 100% ionised)?

- (1) $\text{Al}(\text{OH})_3$
 (2) SrCl_2
 (3) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
 (4) $\text{K}_4[\text{Fe}(\text{CN})_6]$

78. Given below are two statements

Statement-I: During electrolysis of H_2SO_4 solution at higher concentration, SO_4^{2-} is converted to $\text{S}_2\text{O}_8^{2-}$ at anode.

Statement-II: During electrolysis of dilute sulphuric acid water is oxidised at anode.

In the light of above statements choose the correct option among the following

- (1) Both statement-I and statement-II are false
 (2) Statement-I is true but statement-II is false
 (3) Statement-I is false but statement-II is true
 (4) Both statement-I and statement-II are true

Space for Rough Work

Test-5 (Code-A)

Final Test Series for NEET-2024

79. For the 1st order reaction, $A(g) \rightarrow B(g) + 2C(g)$. If the initial pressure is p° and after time t , pressure is p , then the rate constant of reaction is given by
- (1) $k = \frac{2.303}{t} \log \left(\frac{p^\circ}{p} \right)$
 (2) $k = \frac{2.303}{t} \log \left(\frac{p}{p^\circ} \right)$
 (3) $k = \frac{2.303}{t} \log \left(\frac{3p^\circ}{2p^\circ - p} \right)$
 (4) $k = \frac{2.303}{t} \log \left(\frac{2p^\circ}{(3p^\circ - p)} \right)$
80. Following limiting molar conductivities are given as
 $\lambda_m^\circ (\text{Ca(OH)}_2) = x \text{ S cm}^2 \text{ mol}^{-1}$
 $\lambda_m^\circ (\text{Ca(NO}_3)_2) = y \text{ S cm}^2 \text{ mol}^{-1}$
 $\lambda_m^\circ (\text{NH}_4\text{NO}_3) = z \text{ S cm}^2 \text{ mol}^{-1}$
 λ_m° (in $\text{S cm}^2 \text{ mol}^{-1}$) for NH_4OH will be
- (1) $\frac{y}{2} - \frac{x}{2} + z$ (2) $\frac{x}{2} - \frac{y}{2} + z$
 (3) $x - y + \frac{z}{2}$ (4) $x + y - \frac{z}{2}$
81. If the $E_{\text{cell}}^\circ > 0$, then the correct option is
- (1) $\Delta G_{\text{cell}}^\circ < 0$, $K_{\text{eq}} > 1$
 (2) $\Delta G_{\text{cell}}^\circ > 0$, $K_{\text{eq}} < 1$
 (3) $\Delta G_{\text{cell}}^\circ < 0$, $K_{\text{eq}} < 1$
 (4) $\Delta G_{\text{cell}}^\circ > 0$, $K_{\text{eq}} > 1$
82. The mole fraction of the solute in a 1 molal aqueous solution is
- (1) 0.05 (2) 0.5
 (3) 0.17 (4) 0.017
83. The vapour pressure of pure liquids A and B are 100 and 200 mm Hg respectively at 300 K. If the total vapour pressure of the mixture is 175 mmHg then the mole fraction of component A in liquid and vapour phase respectively are
- (1) 0.75 and 0.57 (2) 0.25 and 0.14
 (3) 0.25 and 0.87 (4) 0.25 and 0.75
84. For the reaction, $2A \rightarrow 3B$, the rate of appearance of B with respect to the rate of disappearance of A is
- (1) 2 times (2) $\left(\frac{3}{2} \right)$ times
 (3) 3 times (4) $\left(\frac{2}{3} \right)$ times
85. If rate constant for a decomposition of a reaction is 0.693 min^{-1} and reaction starts with 20 mole of reactant, then rate of reaction after 4 minute will be
- (1) $0.693 \times 5 \text{ mol min}^{-1}$
 (2) $0.693 \times 10 \text{ mol min}^{-1}$
 (3) $0.693 \times 1.25 \text{ mol min}^{-1}$
 (4) $0.693 \times 8 \text{ mol min}^{-1}$
- SECTION-B**
86. The solution which has the highest boiling point is
- (1) 0.3 m aqueous CaCl_2
 (2) 0.2 m aqueous NaCl
 (3) 0.1 m aqueous $\text{K}_4[\text{Fe(CN)}_6]$
 (4) 0.4 m aqueous urea
87. Dissociation constant of 10^{-2} M weak acid (HA) having molar conductance $24 \text{ S cm}^2 \text{ mol}^{-1}$ will be (given: Λ_m° of HA is $480 \text{ S cm}^2 \text{ mol}^{-1}$)
- (1) 5×10^{-5} (2) 6.25×10^{-6}
 (3) 2.5×10^{-5} (4) 5.25×10^{-4}

Space for Rough Work

88. Consider the following statements.

Statement-I: The conductivity of an electrolytic solutions increases on dilution.

Statement-II: The molar conductivity of an electrolytic solution increases on dilution.

In the light of above statements choose the correct option among the following

- (1) Both statement-I and statement-II are true
- (2) Both statement-I and statement-II are false
- (3) Statement-I is true but statement-II is false
- (4) Statement-I is false but statement-II is true

89. A reaction is first order with respect to a gaseous reactant A and second order with respect to a gaseous reactant B. If volume of the mixture is doubled, the rate of reaction becomes

- (1) $\frac{1}{8}$ th times (2) 8 times
- (3) $\frac{1}{4}$ th times (4) 4 times

90. Consider the following statements.

Statement I: The process of flow of solvent through the semi-permeable membrane from pure solvent to the solution is called osmosis.

Statement II: The process of osmosis can be reversed if a pressure higher than the osmotic pressure is applied on the solution side.

In the light of the above statements, identify the correct option.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is incorrect but statement II is correct
- (4) Statement I is correct but statement II is incorrect

91. Osmotic pressure of 20% w/v $\text{Ca}(\text{NO}_3)_2$ solution at 0°C will be (assuming 100% ionisation)

- (1) 27 atm (2) 82 atm
- (3) 50 atm (4) 92 atm

92. Given below are two statements : One labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): Solubility of gases in liquid decreases with rise in temperature.

Reason (R): Solubility of gases in liquid is an exothermic process so solubility decreases with rise in temperature, according to Le-chatelier's principle.

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both Assertion & Reason are true and the Reason is the correct explanation of the Assertion
 - (2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion
 - (3) Assertion is true statement but Reason is false
 - (4) Both Assertion and Reason are false statements
93. The value of resistivity of 0.1 M solution of an electrolyte is $77 \Omega \text{ cm}$. The molar conductivity of the solution is nearly
- (1) $158 \text{ S cm}^2 \text{ mol}^{-1}$ (2) $130 \text{ S cm}^2 \text{ mol}^{-1}$
 - (3) $105 \text{ S cm}^2 \text{ mol}^{-1}$ (4) $50 \text{ S cm}^2 \text{ mol}^{-1}$
94. In a first order reaction $\text{A(g)} \rightarrow 2\text{B(g)}$ the total pressure changes from 100 mm Hg to 150 mm Hg in 10 minutes from the start of the reaction. The rate of appearance of B is
- (1) 5 mm Hg min^{-1} (2) $50 \text{ mm Hg min}^{-1}$
 - (3) $100 \text{ mm Hg min}^{-1}$ (4) $10 \text{ mm Hg min}^{-1}$

Test-5 (Code-A)

95. Cryoscopic constant for a solvent is given by
(where, M_1 = molar mass of solvent,
 T_f = freezing point of pure solvent,
 $\Delta_{fus}H$ = enthalpy of fusion)

$$(1) K_f = \frac{R \times M_1 \times T_f}{1000 \times \Delta_{fus}H}$$

$$(2) K_f = \frac{R \times M_1 \times T_f^2}{1000 \times \Delta_{fus}H}$$

$$(3) K_f = \frac{M_1 \times T_f}{1000 \times R \times \Delta_{fus}H}$$

$$(4) K_f = \frac{1000 \times M_1 \times T_f^2}{R \times \Delta_{fus}H}$$

96. A solution contains 64 g of methyl alcohol and 36 g of H_2O . The mole fraction of methyl alcohol in the solution is

- (1) 0.7 (2) 0.5
(3) 0.9 (4) 0.2

97. In a first order reaction for every $10^\circ C$ rise of temperature, the rate becomes doubled. If the temperature is increased from $40^\circ C$ to $80^\circ C$, the rate of the reaction will become approximately

- (1) 4 times (2) 16 times
(3) 8 times (4) 32 times

98. Standard reduction potentials of the half-cell reactions are given below

$$E_{F_2/F^-}^\circ = 2.85 \text{ V}, E_{Cl_2/Cl^-}^\circ = 1.36 \text{ V}$$

$$E_{Br_2/Br^-}^\circ = 1.06 \text{ V}, E_{I_2/I^-}^\circ = 0.53 \text{ V}$$

The weakest oxidising and reducing agents respectively are

- (1) F_2 and I^- (2) I_2 and F^-
(3) Br_2 and Cl^- (4) F_2 and I_2

99. Given below are two statements : One labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): Molarity is a function of temperature.

Reason (R): Molarity contains volume term.

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both (A) and (R) are correct and (R) is not the correct explanation of (A)
(2) (A) is correct but (R) is not correct
(3) (A) is not correct but (R) is correct
(4) Both (A) and (R) are correct but (R) is the correct explanation of (A)
100. Match List-I (Cell notations) with List-II (E° value)

	List-I		List-II
(A)	$Zn(s) Zn^{2+}(aq) Cu^{2+}(aq) Cu(s)$	(i)	0.46 V
(B)	$Ni(s) Ni^{2+}(aq) Cu^{2+}(aq) Cu(s)$	(ii)	1.1 V
(C)	$Cu(s) Cu^{2+}(aq) Ag^+(aq) Ag(s)$	(iii)	0.59 V
(D)	$Zn(s) Zn^{2+}(aq) Ag^+(aq) Ag(s)$	(iv)	1.56 V

Given

$$(E_{Zn^{2+}/Zn}^\circ = -0.76 \text{ V}, E_{Cu^{2+}/Cu}^\circ = 0.34 \text{ V},$$

$$E_{Ni^{2+}/Ni}^\circ = -0.25 \text{ V}, E_{Ag^+/Ag}^\circ = 0.80 \text{ V})$$

Choose the correct answer from the options given below.

- (1) (A)-(ii), (B)-(iii), (C)-(i), (D)-(iv)
(2) (A)-(iii), (B)-(ii), (C)-(iv), (D)-(i)
(3) (A)-(i), (B)-(iii), (C)-(ii), (D)-(iv)
(4) (A)-(iv), (B)-(i), (C)-(iii), (D)-(ii)

Space for Rough Work

BOTANY

SECTION - A

101. Plants like papaya show
(1) Autogamy (2) Xenogamy
(3) Cleistogamy (4) Geitonogamy
102. The junction between ovule and funicle is represented by
(1) Hilum (2) Chalaza
(3) Integument (4) Nucellus
103. The innermost wall layer of anther can be characterised by all of the following features **except**
(1) Dense cytoplasm (2) Multinucleate
(3) High DNA content (4) Haploid
104. Read the following **assertion (A)** and **reason (R)** and select the **correct** option.
Assertion (A): Cultivation of hybrids has tremendously increased productivity.
Reason (R): Hybrid plants maintain its hybrid characters in their progeny generation after generation.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false
105. Select the **odd** one w.r.t. characteristics of maize plant.
(1) Feathery stigma
(2) Light weighted pollen
(3) Well-exposed stamen
(4) Nectaries
106. Triple fusion involves
(1) Syngamy and double fertilisation
(2) Fusion of one male gamete with one egg
(3) Fusion of one male gamete with two polar nuclei
(4) Fusion of two male gametes with central cell
107. Which of the following statement is **incorrect**?
(1) In pollen grain vegetative cell is larger than generative cell
(2) Endothecium nourishes the developing pollen grains
(3) Intine is made up of cellulose and pectin
(4) Hard outer layer of pollen is called exine
108. An organic substance that can withstand high temperature and cannot be degraded by any enzyme is found in
(1) Tapetal cells of anther wall
(2) Cytoplasm of central cell
(3) Outer layer of sporoderm of pollen grain
(4) Cytoplasm of pollen grain
109. Read the following statements and choose the **correct** option.
Statement (A): Egg apparatus is a 3-celled structure.
Statement (B): Egg apparatus consists of two synergids and one egg cell.
(1) Both statements (A) and (B) are correct
(2) Both statements (A) and (B) are incorrect
(3) Only statement (A) is correct
(4) Only statement (B) is correct
110. How many male gametes are formed by the generative cell of the pollen grain?
(1) Three
(2) Four
(3) Two
(4) Seven

Test-5 (Code-A)

Final Test Series for NEET-2024

111. Read the following statements and select the **correct** option.
Assertion (A): Ovules generally differentiate a single megaspore mother cell (MMC) in the micropylar region of the nucellus.
Reason (R): In majority of flowering plants, monosporic embryo sac development is observed.
 (1) (A) is true but (R) is false
 (2) (A) is false but (R) is true
 (3) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (4) Both (A) and (R) are true but (R) is not the correct explanation of (A)
112. If the number of chromosomes in the secondary nucleus of the *Zea mays* is 20, what is the number of chromosomes in PEC, polar nuclei, coleorhiza and aleurone layer respectively?
 (1) 60, 20, 60 and 40 (2) 30, 10, 10 and 30
 (3) 30, 10, 20 and 20 (4) 30, 10, 20 and 30
113. In geitonogamy
 (1) Transfer of pollen grains occurs within the same flower
 (2) Transfer of pollen grain occurs from the anther of a flower to stigma of another flower of same plant
 (3) No agents are required for pollination
 (4) Always genetically dissimilar type of offsprings are produced
114. Some species of Asteraceae and grasses produce seeds without fertilisation by a special mechanism called
 (1) Apomixis
 (2) Double fertilisation
 (3) Parthenocarpy
 (4) Triple fusion
115. A typical angiospermic flower consists of
 (1) Four fertile whorls
 (2) One non-essential sterile and three essential fertile whorls
 (3) Two non-essential sterile and two essential fertile whorls
 (4) Three essential sterile and one non-essential fertile whorl
116. Co-operative cells in female gametophyte of an angiosperm are characterised by the presence of
 (1) Filiform apparatus
 (2) More than one nucleus
 (3) Photosynthetic pigments
 (4) Chitinous cell wall
117. An embryo directly developed from the cell of the nucellus or integument of a flowering plant would be
 (1) Haploid (2) Polyploid
 (3) Diploid (4) Triploid
118. Select the **odd** one w.r.t. events of pollen-pistil interaction.
 (1) Pollen deposition on the stigma
 (2) Entry of pollen tube into ovule
 (3) Transfer of pollen grains on stigma
 (4) Pollen tube growth in stigma
119. Synthesis of hormone (A) in plants is stimulated by drought, waterlogging and other adverse environmental conditions.
 Select the **correct** feature for hormone (A).
 (1) It helps in ripening of fruit
 (2) It act as an antagonist to GA in terms of dormancy
 (3) It also counteracts apical dominance
 (4) It helps in xylem differentiation
120. Cytokinin and auxin shows their synergistic effect on
 (1) Cell division
 (2) Apical dominance
 (3) Internode elongation
 (4) Promoting dormancy

Space for Rough Work

121. Read the following statements and choose the **correct** option.

Statement (A): All aquatic plants use water for pollination.

Statement (B): Pollination assures the transfer of the right type of pollen.

- (1) Only statement (A) is correct
 (2) Only statement (B) is correct
 (3) Both statements (A) and (B) are correct
 (4) Both statements (A) and (B) are incorrect
122. Select the plant that is pollinated by *Pronuba* moth
- (1) *Ficus* (2) *Hydrilla*
 (3) *Yucca* (4) *Zostera*
123. Match the items in column I with those of column II and choose the **correct** option.

	Column I		Column II
a.	Auxin	(i)	Seed germination
b.	Cytokinin	(ii)	Induces dormancy
c.	Gibberellin	(iii)	Adventitious shoot formation
d.	Absciscic acid	(iv)	Present in root and shoot apices

- (1) a-(i), b-(ii), c-(iii), d-(iv)
 (2) a-(iv), b-(iii), c-(i), d-(ii)
 (3) a-(iv), b-(iii), c-(ii), d-(i)
 (4) a-(iii), b-(iv), c-(i), d-(ii)
124. Precursor of absciscic acid is
- (1) Tryptophan (2) Methionine
 (3) Violaxanthin (4) Acetyl-CoA
125. Identify the following statements as **true (T)** or **false (F)** and choose the option accordingly.
- (A) Development is the sum of two processes growth and differentiation
 (B) Generally, growth is accompanied by anabolic processes only

(C) In arithmetic growth all daughter cell continue to divide

(D) Growth in plants is generally indeterminate

	A	B	C	D
(1)	T	T	T	T
(2)	F	F	T	F
(3)	T	F	F	T
(4)	T	F	F	F

126. Most intensively studied form of gibberellins are
- (1) GA_1 (2) GA_2
 (3) GA_3 (4) GA_4
127. All of the followings are related with cytokinin **except**
- (1) Helps to produce new leaves
 (2) Overcoming apical dominance
 (3) Promotes nutrient mobilisation
 (4) Synchronise fruit-set in pineapples
128. The stimulus for vernalisation is perceived by
- (1) Mature stem apex (2) Young leaf
 (3) Mature leaf (4) Roots
129. Secondary growth in a plant occurs due to
- (a) Vascular cambium
 (b) Cork-cambium
 (c) Root-apical meristem
 (d) Lateral meristem
- The **correct** one(s) is/are
- (1) (a) and (c) only
 (2) (a), (b) and (d) only
 (3) (c) only
 (4) (a) and (d) only
130. The plant growth hormone which is **not** acidic in nature is
- (1) Auxin (2) Gibberellin
 (3) Cytokinin (4) ABA

Test-5 (Code-A)

Final Test Series for NEET-2024

131. Read the following statements and state them as true (T) or false (F) and accordingly select the correct option.
- A. Gardeners uses 2,4-D to make weed free lawn.
B. Auxin is used for hedge making.
C. IAA was isolated from coconut milk by Skoog.
- | A | B | C |
|-------|---|---|
| (1) T | F | T |
| (2) F | T | F |
| (3) T | F | F |
| (4) F | F | T |
132. The gaseous PGR can be used for
(1) Initiate shoot formation on root cuttings
(2) Delaying senescence of leaves
(3) Promoting cell division activity
(4) Inducing flowering in mango
133. Combination of which of the given plant hormones can be used to increase yield of pineapple plants?
(1) Auxin, ABA
(2) ABA, Gibberellins
(3) Auxin, Ethylene
(4) Cytokinin, ABA
134. Cotyledons come out of the soil during seed germination of
(1) Gram (2) Rice
(3) Mango (4) Bean
135. Which of the following phytohormones may involve in growth inhibiting as well as growth promoting activities?
(1) Auxin (2) ABA
(3) Cytokinin (4) Ethylene
- SECTION - B**
136. In dicot embryo, the portion of embryonal axis below the level of cotyledon is called
(1) Hypocotyl (2) Epicotyl
(3) Coleoptile (4) Radicle
137. Endosperm is not completely consumed by the developing embryo in
(1) Groundnut (2) Bean
(3) Pea (4) Wheat
138. Select the incorrect one for anther.
(1) Four sided tetragonal structure
(2) Contain pollen sacs filled with megaspore
(3) Terminal bilobed structure
(4) Tetrastorangiate
139. In angiospermic root, the cells proximal to the meristematic zone do not show
(1) Increased vacuolation
(2) Cell enlargement
(3) New cell wall deposition
(4) Capability to divide
140. Which type of growth curve is shown by roots elongating at a constant rate?
(1) Linear (2) Sigmoid
(3) Hyperbola (4) Parabola
141. Select the incorrect statement w.r.t. pollen grains.
(1) Pollen grains are rich in nutrients
(2) Pollen consumption has been claimed to increase the performance of athletes and race horses
(3) Viability of pollen depends on genetic pre disposition not on prevailing humidity
(4) It is possible to store pollen grains of a large number of species for years in liquid nitrogen (-196°C)
142. In which of the following group of plants, female flowers remain submerged in water and pollen grains are released inside the water, to facilitate pollination?
(1) *Valisneria* (2) Water hyacinth
(3) Water lily (4) Sea grasses

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Final Test Series for NEET-2024

Test-5 (Code-A)

143. In which of the following pair of plants multiple ovules are found in an ovary?
 (1) Mango and Wheat
 (2) Paddy and wheat
 (3) Papaya and water melon
 (4) Orchids and Mango
144. Which of the following insects is a most common pollinating agent?
 (1) Wasps (2) Bees
 (3) Moths (4) Butterfly
145. Majority of insect pollinated flowers
 (1) Have large-feathery stigma to trap pollen
 (2) Have tassels which wave in the wind
 (3) Are large, colourful, fragrant and rich in nectar
 (4) Possess non-sticky pollen grain
146. Consider the following statements and choose the **correct** option.
Statement A: In a young anther, group of compactly arranged homogenous cells is called sporogenous tissue.
Statement B: In a typical stamen, the distal end of the filament is attached to the thalamus of the flower.
 (1) Only statement A is correct
 (2) Only statement B is correct
 (3) Both statements A and B are correct
 (4) Both statements A and B are incorrect
147. During differentiation of xylem, a strong, elastic and A secondary cell wall develops which does not allow tracheids to collapse under extreme tension.

Fill in the blank by choosing the **correct** option for A.

- (1) Suberised (2) Lignocellulosic
 (3) Proteinaceous (4) Cellulosic
148. In plants, relative growth rate can be represented by the formula
 (1) $RGR = \text{Total growth per unit time}$
 (2) $RGR = \frac{\text{Final growth}}{\text{Growth per unit time}}$
 (3) $RGR = \frac{\text{Growth per unit time}}{\text{Initial size}} \times 100$
 (4) $RGR = \frac{\text{Growth per unit time}}{\text{Final size}} \times 100$
149. Consider the following statements and choose the **incorrect** one.
 (1) Long-day plants require exposure to light for a period exceeding the critical period
 (2) The site of perception of light/dark duration in plants are shoot apices
 (3) Photoperiodism is the response of plants to changes in relative lengths of light and dark period
 (4) During flowering shoot apices modify into flowering apices
150. Which plant do not show any correlation between exposure to light duration and induction of flowering response?
 (1) Soyabean (2) Tobacco
 (3) Tomato (4) Wheat

ZOOLOGY

SECTION - A

151. Select the **correct** option to complete the analogy w.r.t. oviduct in a human female.
 Part closer to ovary : Infundibulum :: Part closer to uterus : _____
 (1) Fimbriae (2) Ampulla
 (3) Isthmus (4) Fundus

152. Choose the effect which is **not** caused by catecholamines in humans.
 (1) Decreased blood glucose level
 (2) Increased strength of ventricular contraction
 (3) Increased heart beat
 (4) Raising of hair

Space for Rough Work

Test-5 (Code-A)

Final Test Series for NEET-2024

153. In humans, the secretions of which of the following chiefly helps in the lubrication of the penis?
 (1) Seminal vesicles
 (2) Prostate gland
 (3) Testis
 (4) Bulbourethral glands
154. Implantation is embedding of the _____ into the endometrium of uterus.
 Choose the option that fills the blank **correctly** w.r.t. humans.
 (1) Zygote (2) Blastocyst
 (3) Morula (4) Primary oocyte
155. Hormone which regulates pigmentation of skin by dispersion of melanin granules is
 (1) TSH (2) PRL
 (3) FSH (4) MSH
156. Select the **incorrect** statement w.r.t. acromegaly.
 (1) It is a disorder caused due to hyposecretion of hormone secreted from pituitary gland.
 (2) It generally occurs in middle age person.
 (3) It results in severe disfigurement of the face.
 (4) It is hard to diagnose in the early stages and often goes undetected for many years
157. All of the following are the features of hormones, **except**
 (1) They are non-nutrient chemicals
 (2) Released in trace amounts
 (3) Transported through blood to act on distant target cells
 (4) All thyroid hormones are iodothyronine in nature
158. The structure named 'P' present in female ovary forms an acellular membrane called 'X' around it that on contact with a sperm during fertilisation undergoes changes to prevent polyspermy. Identify 'P' and select the **correct** option.
 (1) Primary follicle
 (2) Secondary oocyte
 (3) Primary oocyte
 (4) Secondary follicle
159. Human placenta is **not** a source of
 (1) Human chorionic gonadotropin
 (2) Human placental lactogen
 (3) Progesterone
 (4) GnRH
160. Select the **correct** option w.r.t. middle layer of uterine wall.
 (1) Glandular in nature
 (2) Undergoes contractions during parturition
 (3) Made up of striated and branched muscle fibres
 (4) Sheds off due to sudden fall in progesterone levels
161. Adenohypophysis consists of two portions named 'X' and 'Y'. 'Y' secretes only one hormone called
 (1) ADH (2) MSH
 (3) ACTH (4) TSH
162. In a pregnant lady, deficiency of iodine in her diet may result in baby having all of the following features, **except**
 (1) Low intelligence quotient
 (2) Deaf - mutism
 (3) Mental retardation
 (4) Protrusion of the eyeballs
163. Reproductive cycle in the female primates is controlled by all of the following, **except**
 (1) Oxytocin from the pars nervosa
 (2) Estrogen and progesterone from the primary sex organ
 (3) FSH and LH from gland located in a bony cavity called sella tursica
 (4) GnRH from the basal part of diencephalon
164. Select the **correct** statement.
 (1) High levels of progesterone triggers the ovulation.
 (2) Colostrum contains readymade antigens and nutrients.
 (3) Spermatozoa get nutrition from Sertoli cells.
 (4) The membranous cover of the ovum during ovulation is yolk sac.

Space for Rough Work

165. Complete the analogy and select the **correct** option w.r.t. humans.

Oogonia : Ovary :: Spermatogonia : _____

- (1) Epididymis (2) Vas deferens
(3) Testis (4) Prostate gland

166. Read the following statements and choose the **correct** option.

Statement A : Small amounts of androgenic steroids are secreted by the adrenal medulla which play a role in the growth of axial hair, pubic hair, and facial hair during puberty.

Statement B : Cortisol is involved in maintaining the cardio-vascular system as well as the kidney functions.

- (1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

167. Inhibiting hormones from hypothalamus reach adenohypophysis through

- (1) Nerve fibres
(2) Portal circulatory system
(3) Arterial system
(4) Lymphatic system

168. Choose the **incorrect** statement w.r.t. humans.

- (1) In adults, each testis is oval in shape with a length of about 2 to 3 cm and a width of 4 to 5 cm.
(2) Decline in LH during luteal phase results in degeneration of corpus luteum in non-pregnant women.
(3) The cells of mammary alveoli secrete milk which is sucked out by the infant through lactiferous duct.
(4) The clitoris is a tiny finger-like structure which lies at the upper junction of the two labia minora above the urethral opening.

169. All of the following are correct w.r.t. secretory phase of menstrual cycle, **except**

- (1) Duration is of 14 days in 28 days menstrual cycle
(2) Degeneration of perimetrium

(3) Low levels of gonadotropins

(4) Development of a temporary endocrine gland

170. How many primary spermatocytes are required to form 10 secondary spermatocytes?

- (1) Two (2) Four
(3) Ten (4) Five

171. Match column I and column II w.r.t. embryonic development in humans and select the **correct** option.

	Column I		Column II
a.	Formation of heart	(i)	During 5 th month of gestation
b.	First movement of foetus	(ii)	End of 1 st trimester of pregnancy
c.	Most of organ system are formed	(iii)	By end of 168 days of gestation
d.	Formation of eyelashes	(iv)	After one month of pregnancy

- (1) a-(i), b-(ii), c-(iii), d-(iv)
(2) a-(iv), b-(i), c-(ii), d-(iii)
(3) a-(ii), b-(i), c-(iv), d-(iii)
(4) a-(i), b-(iii), c-(iv), d-(ii)

172. Hormones are involved in

- a. Coordination b. Integration
c. Regulation

Select the **correct** option.

- (1) Only a (2) Only a and b
(3) Only c (4) a, b and c

173. Choose the **correct** match.

(1)	Gastrin	–	Inhibits gastric motility
(2)	Secretin	–	Stimulates secretion of bile juice from gall bladder
(3)	GIP	–	Stimulates secretion of HCl and pepsinogen
(4)	CCK	–	Stimulates the secretion of pancreatic enzymes

Test-5 (Code-A)

Final Test Series for NEET-2024

174. How many of the organ(s)/gland given in the box below consist of a central medullary region surrounded by a cortical region in humans?

Ovary, Adrenal gland, Kidney, Liver, Pancreas

Select the **correct** option.

- (1) Two (2) Three
(3) Five (4) One
175. Comprehend the given statements w.r.t. humans.

Statement (A): Pars distalis produces six trophic hormones.

Statement (B): Pineal gland is located on the ventral side of forebrain.

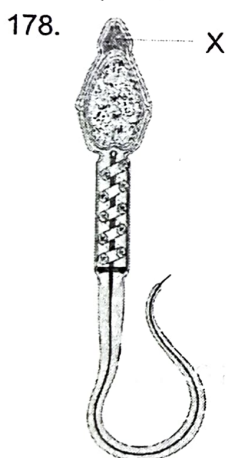
Select the **correct** option.

- (1) Both statements (A) and (B) are correct
(2) Both statements (A) and (B) are incorrect
(3) Only statement (A) is correct
(4) Only statement (B) is correct
176. The glandular tissue of each breast in human females is divided into how many mammary lobes?

- (1) 15-20 (2) 30-40
(3) 20-30 (4) 10-15

177. Choose the hormone which is present only in pregnant women and produced by ovary.

- (1) Progestogens (2) Estrogen
(3) hPL (4) Relaxin



Identify the labelled structure 'X' and select the **correct** option w.r.t. it.

- (1) Filled with enzymes that help in fertilization of ovum
(2) Contains an elongated haploid nucleus
(3) Produces energy for the movement of tail
(4) Lacks plasma membrane around it
179. Parturition is induced by a complex _____ mechanism. The signals for parturition originate from the _____ and the placenta which induce mild uterine contractions called foetal ejection reflex. Choose the **correct** option to fill in the blanks respectively.
- (1) Neural, maternal pituitary
(2) Neuroendocrine, fully developed foetus
(3) Endocrine, foetal pituitary
(4) Neuroendocrine, half developed foetus
180. Select the **correct** pair of second messengers which are generated when hormones interact with membrane bound receptors.
- (1) Cyclic AMP, ATP (2) IP_3 , Ca^{++}
(3) T_3 , Ca^{+2} (4) Ca^{++} , ATP
181. Consider the following features:
- (a) Released by neurohypophysis
(b) Acts on the smooth muscles of uterus and stimulates their contractions during parturition
(c) Acts on the mammary gland for the ejection of milk
- The above given features are associated with
- (1) Progesterone (2) Oxytocin
(3) Estrogen (4) Prolactin

182. **Assertion:** Insulin is a hypoglycemic hormone.
Reason: It stimulates glycogenolysis and inhibits gluconeogenesis.

In the light of the above statements, select the **correct** option.

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) Both (A) and (R) are false
(4) (A) is true but (R) is false

Space for Rough Work

Final Test Series for NEET-2024

Test-5 (Code-A)

183. Thymosin is responsible for
- (1) Differentiation of T-lymphocytes
 - (2) Inhibition of gastric secretions
 - (3) Suppressing the immune response
 - (4) Maturation of the male accessory sex organs
184. Estrogen enters into cells and binds to intracellular receptors instead of membrane-bound receptors because it is a/an
- (1) Steroid hormone
 - (2) Iodothyronine
 - (3) Amino-acid derivative hormone
 - (4) Peptide hormone
185. A peptide hormone released by a composite gland plays a major role in regulation of glucose homeostasis and acts on hepatocytes for enhanced glucose uptake. Identify the hormone and choose the **correct** option.
- (1) Testosterone
 - (2) Glucagon
 - (3) Insulin
 - (4) Vasopressin

SECTION - B

186. In humans, which of the following structures differentiates into endoderm and ectoderm and have the potency to give rise to all the tissues and organs?
- (1) Yolk sac
 - (2) Umbilical cord
 - (3) Inner cell mass
 - (4) Trophoblast
187. The **correct** pathway of transport of sperms passing through a human male body is
- (1) Seminiferous tubules → Rete testis → Vasa efferentia → Epididymis → Vas deferens → Ejaculatory duct → Urethra
 - (2) Seminiferous tubules → Rete testis → Vas deferens → Epididymis → Vasa efferentia → Ejaculatory duct → Urethra
 - (3) Seminiferous tubules → Epididymis → Vas deferens → Rete testis → Vasa efferentia → Ejaculatory duct → Urethra
 - (4) Seminiferous tubules → Epididymis → Vas deferens → Rete testis → Vasa efferentia → Urethra → Ejaculatory duct

188. Choose the structure that acts as temporary endocrine gland and is not present since birth in humans.

- (1) Corpus luteum
- (2) Thymus
- (3) Pineal gland
- (4) Thyroid gland

189. Match Column I with Column II w.r.t. chemical nature of hormones.

Column I

Column II

- | | |
|-------------------|-----------------------|
| a. Endorphins | (i) Protein |
| b. Noradrenaline | (ii) Iodinated amine |
| c. Growth hormone | (iii) Biogenic amines |
| d. Thyroxine | (iv) Neuropeptides |

Select the **correct** option.

	a	b	c	d
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iii)	(iv)	(i)	(ii)
(3)	(iv)	(iii)	(ii)	(i)
(4)	(iv)	(iii)	(i)	(ii)

190. Which of the following hormones is used in transplantation surgery to suppress immune response in humans?

- (1) Cortisol
- (2) Aldosterone
- (3) T₃
- (4) Thymosin

191. Spike in level of which hormone is cause of ovulation?

- (1) hCG
- (2) Prolactin
- (3) LH
- (4) Progesterone

192. Which of the following is indicative of diabetes mellitus?

- (1) Uremia and glycosuria
- (2) Glycosuria and ketonuria
- (3) Uremia and diuresis
- (4) Uremia and ketonuria

Space for Rough Work

Test-5 (Code-A)

Final Test Series for NEET-2024

193. Low levels, of which hormone is associated with tetany?
(1) Thyrocalcitonin (2) Epinephrine
(3) Norepinephrine (4) PTH
194. Choose the **correct** match.
(1) Aldosterone – α -cells of Islets of Langerhans
(2) Cortisol – Mineralocorticoid
(3) Insulin – β -cells of pancreas
(4) Glucagon – Glucocorticoid
195. Sperms undergo biochemical changes in the female reproductive tract which enables them to fertilize an ovum. This process is called
(1) Spermiation
(2) Spermatogenesis
(3) Capacitation
(4) Insemination
196. Which of the following is associated with hormone released from the endocrine gland 'X' that is located on the back side of the gland 'Y' which is located on either side of the trachea?
(1) Regulation of basal metabolic rate
(2) Stimulates reabsorption of Ca^{+2} by the renal tubules
(3) Development of the immune system
(4) Control metabolism of carbohydrates, proteins and fats
197. In humans, male pronucleus fuses with
(1) Pronucleus of primary oocyte
(2) Pronucleus of ootid
(3) Haploid nucleus of second polar body
(4) Haploid nucleus of female germ cell
198. For normal fertility, out of 200-300 million human sperms, at least how many sperms have normal shape and size and must show vigorous motility?
(1) 120-180 million
(2) 48-72 million
(3) 100-150 million
(4) 60-80 million
199. In human females the three phases of the ovarian cycle in correct order is
(1) Menstrual $\rightarrow$ Luteal $\rightarrow$ Ovulatory
(2) Follicular $\rightarrow$ Ovulatory $\rightarrow$ Luteal
(3) Menstrual $\rightarrow$ Proliferative $\rightarrow$ Secretory
(4) Ovulatory $\rightarrow$ Luteal $\rightarrow$ Proliferative
200. The hormone androgens
(1) Are secreted from testis under the influence of FSH
(2) Influence the male sexual behaviour
(3) Produce anabolic effects on only protein metabolism
(4) Are responsible for high pitch of voice in males

□ □ □

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